

Jialin Huang

Software Engineer

linao264590000@gmail.com linkedin.com/in/jlhuangprogramming github.com/jialinhuang00 jialin00.com

Open to relocation: Netherlands / Germany / EU · Mandarin (native), English (professional)

Backend engineer with a frontend foundation, building REST and GraphQL APIs on a multi-tenant cybersecurity SaaS: policy config, RBAC, SAML, Cognito audit. On my own I go deeper: a Go URL shortener that splits OLTP from OLAP, and a RAG pipeline with its own eval harness. I build APIs the client actually wants to consume.

SKILLS

Backend: NestJS, Apollo GraphQL, Golang, Gin, WebSocket, Node.js, GraphQL, SignalR, Firebase Auth, Python/Flask

Security & Domain: RBAC, SAML, OAuth, audit logging, Zero Trust access, multi-tenant SaaS

Infrastructure: PostgreSQL, Docker, Kubernetes, Jenkins, AWS Lambda, AWS Cognito, Google Analytics, Firebase, GCP

Frontend: React, Next.js, Remix, Vue, Canvas, Egret, Tailwind CSS, Emotion, Redux, C#/.NET

EXPERIENCE

MammothCyber

Senior Full Stack Engineer · Cybersecurity · Taipei (Hybrid) · Apr 2025 - Present

Full-stack across the web portal of a multi-tenant enterprise security platform, a Chromium-based secure browser, and Go services; security domain (access control, audit, policy config)

Console Migration to Remix

Migrated console pages onto Remix (React 18, Mantine, TanStack Query/Table, Vite) off an earlier Vue / Next.js stack, page by page with new and old running side by side

Policy Configuration Console

Built the policy-configuration backend in NestJS: rule and content-profile APIs, RBAC-gated, with audit logging and an N+1 fix that collapsed a multi-call save into a single request

Built the rules and content-profile editor in Remix, type-safe end to end with reactive validation

Secure Browser & Extension (Chromium)

Shipped a login / session-visibility feature across every client of a Chromium-based enterprise browser, extending each platform's native bridge in C++ (Android / Desktop) and Objective-C++ (iOS), plus a Go fetch backend and a React UI

Built the login dialog in the companion Chrome extension (React / Redux), surfacing device info and recent failed logins

Identity, Access & Audit

Worked across the auth stack (Go + Node): added tenant-level forced re-authentication (SAML), and threaded an OS / device-posture field across the login-policy service, the Cognito audit pipeline, and the browser and extension clients

Integrated Google / Apple social login into Cognito on top of the existing SAML, reconciling the OAuth scopes for external IdPs and reworking several Cognito Lambda triggers, including the PreTokenGeneration hook that maps each social user to its tenant

Integrated credentials and storage across AWS, Azure, and GCP (presigned URLs, Key Vault signing)

Platform & Build Infrastructure

Drove a monorepo dependency cleanup on npm: root overrides and targeted upgrades (Express v5, Apollo Server v5, ESLint v9) cut vulnerabilities 74 → 13 (-82%), clearing every high & critical, and stray local packages 22 → 0

Migrated the monorepo to pnpm workspaces with a staged rollout (shamefully-hoist bridge, declare every phantom dep, flip to strict isolation), cutting installs 3-10x and making per-worktree installs cheap so parallel AI agents can each build in their own git worktree

Cut two Docker images ~80% with bundling and pnpm deploy --prod: Remix web app 1.88 GB → 264 MB, NestJS API 1.95 GB → 490 MB; compressed registry size ~540 MB → 89 MB (-84%) and 628 MB → 180 MB (-71%)

Bunastar

Frontend Engineer · Gaming · Taipei (Hybrid) · Nov 2024 - Apr 2025

Real-time multiplayer card game for the Southeast Asian market; custom Canvas engine and WebSocket sync

Southeast Asian Gaming Platform - Pok Deng

Card game logic: betting, dealing, scoring, payout

Real-time card sync over WebSocket

Canvas rendering engine, optimized with sprite sheets

Cross-domain communication via postMessage API

Confetti

Full Stack Engineer · Gaming · Taipei (Hybrid) · Dec 2022 - Oct 2024

confetti.studio

Y2K landing page

Cashing Platform

Built and iterated payment and cash-out flows across multiple product lines

SCMP

Frontend Engineer · Media · Hong Kong (Remote) · Oct 2021 - Nov 2022

South China Morning Post: high-traffic international news platform, delivered fully remote

scmp.com

Improved Core Web Vitals (LCP, CLS), Google's SEO ranking signals, on scmp.com, a high-traffic international news site: fixed the homepage hero's layout shift, lazy-loaded below-fold images, and tuned SSR data prefetch

Maintained the CMS-driven content rendering pipeline (Node.js, GraphQL)

Built the site-wide tracking plan across sections: GA events through the GTM dataLayer for traffic, and Mixpanel via SDK keyed on the reader's uuid for per-user product behavior

Applied GDPR right-to-erasure in the comment system: surfaced a gdpr_erased flag through the GraphQL layer so erased users' names render as Anonymous

Futuresai

Frontend Engineer · Finance · Taipei (Hybrid) · Nov 2020 - Oct 2021

futures-ai.com

Built a subscription-billing system with multiple tiers

Live trading strategy articles with graphical analysis

Algorithm-driven financial insights

Trading Dashboard + Admin + Auth

Monorepo rewrite with SignalR for real-time trading-data streaming

SMS and LINE push notifications

Firebase auth middleware with custom membership logic

Real-time products in a financial-trading domain

Dentalltw

Frontend Engineer · Healthcare · Taipei (Hybrid) · Jul 2018 - Nov 2020

Built a 0-to-1 clinic product: a desktop app for Taiwan and a React web app for the Vietnam market

Dentall Clinic Management

Built the desktop clinic app (UWP, C#/.NET) for registration, appointments, billing, and prescriptions, integrated with Taiwan's NHI (National Health Insurance) system

Built Taiwan's first SVG-based dental chart for procedure documentation

Vietnam Market Expansion

Built the Vietnam web app (React, Redux): the clinic software localized, with multi-language support and SMS

CI/CD Pipeline

Set up CI/CD to build and push Docker images to GCR, using a Go service and GitLab scripts

SAVY

Frontend Intern · Taipei (Hybrid) · Apr 2018 - Jul 2018

Community Help Board

A SPA for posting and browsing local community help requests

PROJECTS

URL Shortener · Go, Gin, PostgreSQL, Redis, Kafka, ClickHouse

One redirect touches five systems: PostgreSQL for link mappings, Redis for live counters, Kafka for the click pipeline, ClickHouse for analytics, and a Go/Gin service tying them together.

Clicks flow through Kafka in 500-event / 2s batches across 4 writer goroutines, so analytics writes never block a redirect

Write-behind counter: redirects INCR in Redis, a 5s worker drains to PostgreSQL; reads merge the durable total with the live delta

Live click dashboard over Server-Sent Events: a pub/sub hub fans out each redirect to all clients, plus a per-link 2s stream merging the Postgres total with the Redis delta

Hand-rolled a Redis-bitset Bloom filter to short-circuit lookups for keys that do not exist

221b · Python, ChromaDB, sentence-transformers, MCP

My outie and my innie in one searchable self. Severance, but merged instead of split. Ask what my brain has shipped, how I think, or "what would I do?", and it answers as me.

One corpus of my blog, my Claude sessions, and my inner config (soul, persona, style).

Chinese question → embeddings retrieve → reranker reorders → my English writing.

Answers "if you were me, how would you decide?" from my values and past calls.

EDUCATION

National Chengchi University (NCCU) · B.A. in Economics